

DATA
SET

DS1509

6. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

- a. Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.
- b. Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

Pair	1	2	3	4	5	6	7
Population 1 - Population 2	0.1	0.7	0.3	-0.1	0.5	0.2	0.5
Rank	1.5	7	4	1.5	5.5	3	5.5

$$T^+ = 1.5 + 7 + 4 + 5.5 + 3 + 5.5 = 26.5$$

$$T^- = 1.5 \quad T = \min(T^+, T^-) = T^- = 1.5$$

H_0 : the two population distribution are the same, H_a : the two population distribution are different.

$$\alpha = 0.1$$

$$RR: \{T \mid T \leq 4\}$$

$T = 1.5 \leq 4 \Rightarrow \text{Reject } H_0 \Rightarrow \text{We have sufficient evidence to say that two population distribution are different.}$